Customer Behavior Analysis Report (Detailed Summary)

1. Introduction

This report analyzes customer behavior to understand purchasing patterns, satisfaction, recommendation effectiveness, and product associations using Apriori and clustering techniques.

2. Data Preparation

Dataset was cleaned, categorical variables standardized, and ordinal variables converted to numeric. No major missing value issues remained.

3. Descriptive Analysis

Average age ≈ 34 years. Gender distribution is balanced.

Purchase frequency is evenly distributed, showing varied customer behavior.

Popular categories: High Demand for :- Beauty & Personal Care, Clothing & Fashion, Home & Kitchen. Customers prefer lifestyle and daily-use products

Browsing frequency shows high engagement but includes low-frequency users.

Cart abandonment is common, indicating conversion issues.

Shopping satisfaction ≈ 3 (moderate), indicating scope for improvement.

4. Segmentation & Clustering

Segments: At-Risk Customers (534), Occasional Shoppers (154), Frequent Buyers (112).

K-Means clustering (k=3) confirmed behavioral groupings.

Frequent Buyers: high satisfaction and purchase frequency.

At-Risk Customers: low satisfaction and engagement. Large portion falls into it.

5. Recommendation Insights

Recommendation helpfulness vs satisfaction correlation ≈ 0.02 → very weak.

Increasing recommendation frequency does not increase trust.

Users perceive recommendations as moderately helpful (value ≈ 2).

6. Review Insights

Review reliability and helpfulness show near-zero correlation with rating accuracy.

Customers rely more on personal experience than reviews.

7. Apriori Analysis Insights

Strong association rules identified:

- (Groceries + Others) → Clothing (confidence 0.65, lift 1.23)
- (Others + Home) → Clothing (confidence 0.61, lift 1.15)
- (Groceries + Home) → Clothing (confidence 0.58, lift 1.10)

Clothing acts as a central cross-selling category.

8. Visual Insights

Bar charts show popularity of lifestyle categories.

Browsing distribution indicates moderate to high engagement.

Pie chart shows majority users are moderately satisfied.

Heatmap confirms weak relationship between recommendation helpfulness and satisfaction.

9. Business Recommendations

Improve personalization using user behavior data.

Use Apriori rules for cross-selling recommendations.

Reduce irrelevant recommendations.

Add explainability (why recommended).

Target At-Risk customers with offers.

Improve product quality and delivery experience.

10. Conclusion

Customer satisfaction is moderate and recommendation systems currently have limited impact.

Data-driven personalization and association-based recommendations can significantly improve engagement and conversion.

Key Takeaway:

Improving recommendation quality—not quantity—is critical for enhancing customer satisfaction and business performance.

Video Reference :-
https://drive.google.com/drive/u/0/folders/1s70AZhGcWHKNMKJVvgYInirOVnuJ2MLd